



Hyperloop Transportation

The Hyperloop concept looks optimistic with the project already underway in California.
<http://dgit.in/TravelFast>



US Octopus

Find out what the Octopus of U.S. National Reconnaissance Office entails.
<http://dgit.in/OctoUS>

Did we really just find a long-lost giant planet?

A sudden revelation struck recently, with 'conclusive' evidence of a ninth planet orbiting the Sun at the outer fringe of our solar system. The scientists behind this believe that the planet may be about four times larger than Earth, and almost 10 times its mass. How did we manage to miss it until now?

Souvik Das
souvik.das@digit.in

In 2003, Caltech's leading astronomer Mike Brown spotted Sedna, a dwarf planet at the outer edges of the solar system. The planet had an elongated elliptical orbit, making its closest approach to the Sun at 76 AU (Astronomical Unit, where 1 AU = distance between the Earth and the Sun, that is 150 million kilometres), and the farthest it gets is 937 AU. Through the next 12 years, five more such planets were spotted, with similar orbits, giving rise to a speculation among scientists that there is a bigger entity out there possessing much greater gravitational influence.

After much chalkboarding and numerous astronomical calculations, the scientists came to a consensus on the presence of Planet X — a 9th 'planet', large enough to assert its gravitational dominance to clear out debris from its neighbourhood and hold dwarf planets in orbit. This is quite the standard sce-

nario for previous 'Planet X discoveries'. Ever since discrepancies were spotted in Uranus' orbit, there was debate on the existence of a planet large enough to create such a distinct impact. The mid-1850s finally saw the discovery of Neptune — a startling find at that point of time.

Percival Lowell's renewed quest for a farther Planet X ended with the discovery of Pluto, in 1930. However, Pluto was a dwarf — a small, rocky, icy planet with its orbit tilted out of the plane of the solar system. In comparison to the gaseous giants that precede it in our solar system, Pluto is a very odd celestial being. Perhaps, it is Pluto's constitution that made scientists think about its actual status at the outer edge of our solar family.

Brown, in 2005, spotted Eris. This object, like Pluto and Sedna, was 'tiny', far out into space, and had a similar setup like the two dwarfs. This suggested Brown that many other such worlds may exist at the outer edge of our solar system. When evidence led to the

spotting six such planets with a similar orbit system, Brown and his colleague Konstantin Batygin started discussing the reason behind such occurrences. They both came to a conclusion — something that astronomers have been trying hard to state for near centuries now, but have only ended up in blind alleys. Brown and Batygin exclaimed that it had to be a planet causing such a setup to occur and sustain itself.

Brown and Batygin have been cautious in their approach, taking time to study computer-simulated models, possible orbits, presence of nearby or passing stars, and came up with a theory. There is an actual ninth planet at the outer edge of the solar system, large enough to influence six other smaller bodies with its gravitational force. Since that is quite a significant amount of force, no dwarf planet can do that, as a result of which the mass and size of the planet were guessed. Adding to this, the planet has not been seen yet, meaning that it must have an orbit that is irregular, or massive, or in a different direction, or all of these at the same time.

The scientists presume, during the Solar System's young days of about 4 billion years ago, most of the gaseous planets on the outer edge of the Solar System had rocky cores. Had Planet X remained in the inner circles of the solar system, it would have possibly gathered up ice and gases to become a giant, like Jupiter. However, the space was so congested that all the planets could not have expanded to the fullest, and Planet X, with its outer orbit, got eliminated. At that time, the Solar System had a gaseous nebula on its outer edge, that deviated and slowed the planet down in its natural course, which may have given it the irregular elliptical